

## MODULE V GREEN FUELS

### What are Biofuels?

Any hydrocarbon fuel that is produced from an organic matter (living or once living material) in a short period of time (days, weeks, or even months) is considered a biofuel.

Biofuels are energy sources made from recently grown biomass (plant or animal matter). Biofuels have been around for a long time

Biofuels are a renewable resource because they are continually replenished

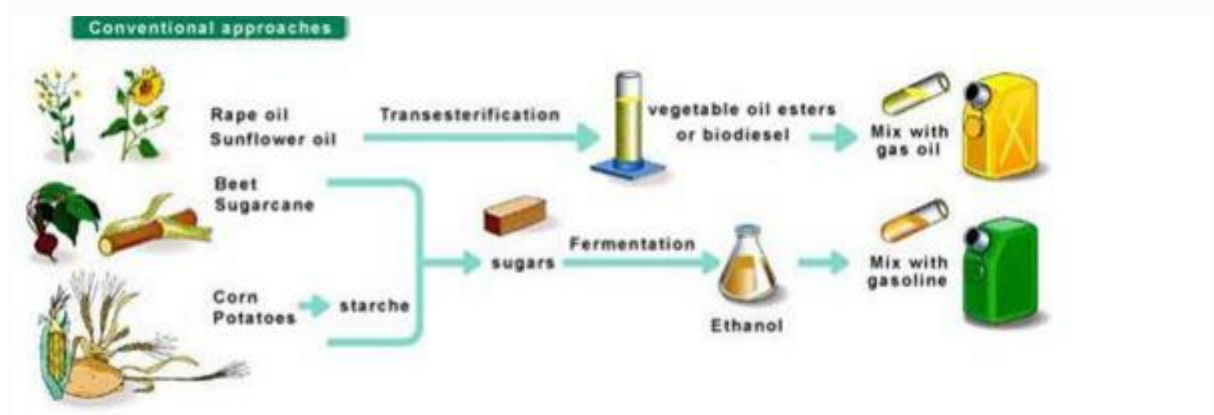
Biofuels may be solid, liquid or gaseous in nature.

1. Solid: Wood, dried plant material, and manure
2. Liquid: Bioethanol and Biodiesel
3. Gaseous: Biogas

### Classification of Biofuels:

**1st generation biofuels** First generation biofuels, also known as conventional biofuels, are made from sugar, starch or vegetable oil. First generation biofuels are produced through well-understood technologies and processes, like fermentation, distillation and transesterification. These processes have been used for hundreds of years in many uses, such as making alcohol. Sugars and starches are fermented to produce primarily ethanol, and in smaller quantities, butanol and propanol. Ethanol has one-third of the energy density of gasoline, but is currently used in many countries, including the United States, as an additive to gasoline. A benefit of ethanol is that it burns cleaner than gasoline and therefore produces less greenhouse gases. Another 1st generation biofuel, called biodiesel, is produced when plant oil or animal fat goes through a process called transesterification. This process involves exposing oils with an alcohol such as methanol in the presence of a catalyst. The distillation process involves separating the main product from any of the byproducts of the reactions. Biodiesel can be used in place of petroleum diesel in many diesel engines or in a mixture of the two. The Figure 1 shows the process of how 1st generation biofuels are made. First generation biofuels symbolize a step towards energy independence and weaning off fossil fuels for energy demands. These biofuels also support agricultural industries and rural communities through increased demand for crops. This being said, 1st generation biofuels also have several disadvantages. They pose a threat to food prices since the biomass used are food crops such as corn and sugar beet. First

generation biofuel production has contributed to recent increases in world prices for food and animal feeds. They also have the potential to have a negative impact on biodiversity and competition for water in some regions. Additionally, biomass for first generation biofuels requires lots of land to grow, and this provides only limited greenhouse gases reduction. They also only provide a small benefit over fossil fuels in regards to greenhouse gases since they still require high amounts of energy to grow, collect, and process. Current production practices use fossil fuels for power. First generation biofuels are also a more expensive option than gasoline, making it economically unfavorable. Finally, biodiesel almost always comes from recycled oils from restaurants, as opposed to virgin oils, so the supply is limited by restaurants' oil use.



**2nd generation biofuels** The biomass sources for 2nd generation biofuels include wood, organic waste, food waste and specific biomass crops. Fast growing trees such as poplar trees need to undergo a pretreatment step, which is series of chemical reactions that break down lignin, the “glue” that holds plants together, in order to make fuel. This pretreatment step involves thermochemical or biochemical reactions that unlock the sugars embedded in fibers of the plant. After this step, the process to generate plant ethanol resembles that of 1st generation ethanol production. Additionally, straw and other forest residues can go through a thermochemical step that produces syngas (a mixture of carbon monoxide, hydrogen and other hydrocarbons). Hydrogen can be used as a fuel and the other hydrocarbons can be

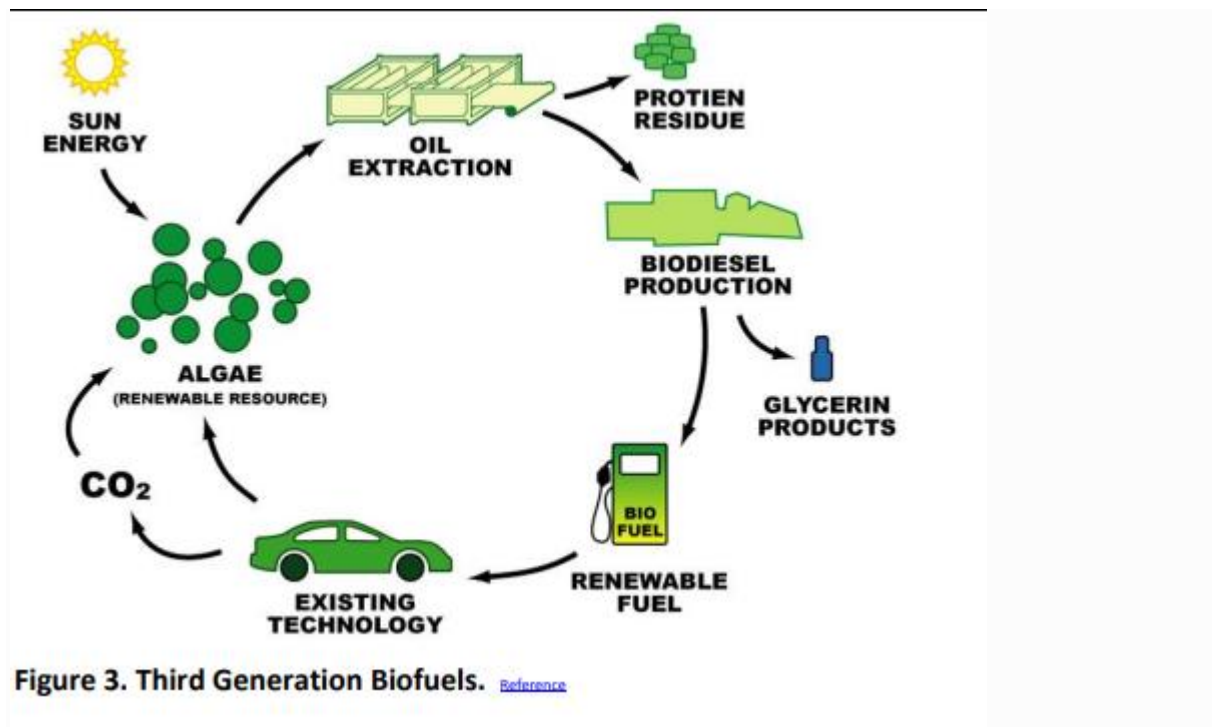
used as additives to gas oil. The Figure 2 shows how 2nd generation biofuels are produced



Second generation biofuels address many issues associated with 1st generation biofuels. They don't compete between fuels and food crops since they come from distinct biomass. Second generation biofuels also generate higher energy yields per acre than 1st generation fuels. They allow for use of poorer quality land where food crops may not be able to grow. The technology is fairly immature, so it still has potential of cost reductions and increased production efficiency as scientific advances occur. However, some biomasses for second-generation biofuels still compete with land use since some of the biomass grows in the same climate as food crops. This leaves farmers and policy makers with the hard decision of which crop to grow. Cellulosic sources that grow alongside food crops could be used for biomass, such as corn stover (leaves, stalk, and stem of corn). However, this would take away too many nutrients from the soil and would need to be replenished through fertilizer. In addition, the process to produce 2nd generation fuels is more elaborate than 1st generation biofuels because it requires pretreating the biomass to release the trapped sugars. This requires more energy and materials.

### 3rd generation biofuels

Third generation biofuels use specially engineered crops such as algae as the energy source. These algae are grown and harvested to extract oil within them. The oil can then be converted into biodiesel through a similar process as 1st generation biofuels, or it can be refined into other fuels as replacements to petroleum-based fuels. The Figure 3 shows the general steps of how 3rd generation biofuels are produced.



Third generation biofuels are more energy dense than 1st and 2nd generation biofuels per area of harvest. They are cultured as low-cost, high-energy, and completely renewable sources of energy. Algae are advantageous in that it can grow in areas unsuitable for 1st and 2nd generation crops, which would relieve stress on water and arable land used. It can be grown using sewage, wastewater, and saltwater, such as oceans or salt lakes. Because of this, there wouldn't be a need to use water that would otherwise be used for human consumption. However, further research still needs to be done to further the extraction process in order to make it financially competitive to petrodiesel and other petroleum-based fuels.

| Sr. No. | Biofuels                                                                                               | Fossil fuels                                                                         |
|---------|--------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| 1.      | Biofuels are produced from <u>present-day biomass</u> .                                                | Fossil fuels are product of biomass present <u>millions of years ago</u> .           |
| 2.      | Biofuels are <u>renewable energy</u> sources.                                                          | Fossil fuels are <u>non-renewable energy</u> sources.                                |
| 3.      | Biofuels release net <u>zero carbon dioxide</u> (CO <sub>2</sub> neutral process) in the environment.. | Fossil fuels increase <u>the concentration of carbon dioxide</u> in the environment. |
| 4.      | Biofuels are <u>sustainable sources of energy</u> .                                                    | Fossil fuels are <u>non-sustainable sources of energy</u> .                          |
| 5.      | <u>Energy generation</u> is low                                                                        | Energy generation is <u>comparatively high</u> .                                     |
| 6.      | Require more investment for their production and are costly.                                           | Comparatively less investment and less cost to biofuel.                              |

### biofuels also have several challenges

- Limited availability: Biofuels are only produced from certain feedstocks.
- Food versus fuel debate: First-generation biofuels, such as bioethanol, use food crops.
- Greenhouse gas emissions: First-generation biofuels are associated with greenhouse gas emissions.
- Fuel costs: Biofuels can be expensive to produce.

| Challenge                                                                                                                                                                                                                                                                  |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Feed stock</b><br>Collection network<br>Storage facilities<br>Food-fuel competition                                                                                                                                                                                     |
| <b>Technology</b><br>Pretreatment<br>Enzyme production<br>Efficiency improvement<br>Technology cost<br>Production of value added co-products                                                                                                                               |
| <b>Policy</b><br>Land use change<br>Fund for research and development<br>Pilot scale demonstration<br>Commercial scale deployment<br>Policy for biofuels<br>Procurement of subsidies<br>on biofuels production<br>Tax credits on production<br>and utilization of biofuels |

### Government of India initiatives to promote the use of Biofuels:

Since 2014, the Government of India has taken a number of initiatives to increase blending of biofuels.

1. The major interventions include administrative price mechanism for ethanol, simplifying the procurement procedures of OMCs, amending the provisions of Industries (Development & Regulation) Act, 1951 and enabling lignocellulosic route for ethanol procurement.

2. The Government approved the National Policy on Biofuels-2018 in June 2018. The policy has the objective of reaching 20% ethanol-blending and 5% biodiesel-blending by the year 2030.
  - Among other things, the policy expands the scope of feedstock for ethanol production and has provided for incentives for production of advanced biofuels.
3. The Government has also increased the price of C-heavy molasses-based ethano

### **Salient features of the National Biofuels policy 2018**

- Categorisation of biofuels to enable extension of appropriate financial and fiscal incentives under each category. The two main categories are:
  - Basic Biofuels- First Generation (1G) bioethanol & biodiesel
  - Advanced Biofuels – Second Generation (2G) ethanol, Municipal Solid Waste (MSW) to drop-in fuels, third Generation (3G) biofuels, bio-CNG etc.
- Expands the scope of raw material for ethanol production by allowing use of Sugarcane Juice, Sugar containing materials like Sugar Beet, Sweet Sorghum, Starch containing materials like Corn, Cassava, Damaged food grains like wheat, broken rice, Rotten Potatoes, unfit for human consumption for ethanol production.
- Allows use of surplus food grains for production of ethanol for blending with petrol to ensure appropriate price to farmers during surplus. However, it needs the approval of the National Biofuel Coordination Committee.
- Thrust on Advanced Biofuels: Viability gap funding scheme for 2G ethanol Bio refineries of Rs.5000 crore in 6 years in addition to additional tax incentives and higher purchase price as compared to 1G biofuels.
- Encourages setting up of supply chain mechanisms for biodiesel production from non-edible oilseeds, used Cooking Oil, short gestation crops.
- Synergising efforts by capturing the roles and responsibilities of all the concerned Ministries/Departments with respect to biofuels in the policy document itself.

### **Potential Benefits**

#### **Potential Benefits of Bio Fuels**

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## Benefits

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### **Energy Security**

Domestic energy source  
Locally distributed  
Well connected supply-demand chain  
Higher reliability

### **Economic stability**

Price stability  
Employment generation  
Rural development  
Reduce inter-fuels competition  
Reduce demand-supply gap  
Open new industrial dimensions  
Control on monopoly of fossil rich states

### **Environmental gains**

Better waste utilization  
Reduce local pollution  
Reduce GHGs emission from energy consumption  
Reduction in landfill sites

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- **Reduce Import Dependency:** The large-scale production of biofuels would reduce import dependency on crude oil and save forex.
- **Cleaner Environment:** By reducing crop burning & conversion of agricultural residues/wastes to biofuels there will be reduction in GHGs emissions and other particulate matters.
- **Municipal Solid Waste Management:** It is estimated that, annually around 62 MMT of Municipal Solid Waste gets generated in India. The policy promotes conversion of waste/plastic, MSW to drop in fuels (hydrocarbon fuels from solid waste).
- **Infrastructural Investment in Rural Areas:** addition of 2G bio refineries across the Country will spur infrastructural investment in the rural areas.
- **Employment Generation:** the establishment of bio-refineries would create jobs in Plant Operations, Village Level Entrepreneurs and Supply Chain Management.
- **Additional Income to Farmers:** Farmers can capitalize on agricultural residues /waste which otherwise are burnt by them. They can sell their



surplus output to ethanol making units when price dump, thus, ensuring appropriate price.

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- The Indian government launched the Ethanol Blended Petrol (EBP) initiative in January 2013, which contains a 5% ethanol blend in petrol. To expand biofuel (bioethanol and biodiesel) consumption and reduce India's present dependency on fossil fuels, the initiative placed a heavy emphasis on India's agricultural and industrial capabilities. By 2025, a 20% ethanol blend may save the government Rs 30,000 crore in foreign currency, promote energy security, reduce carbon emissions, improve air quality, increase self-reliance, utilize damaged foodgrains, increase farmers' earnings, create jobs, and expand investment prospects. To meet the demand for its 20% Fuel Blending Program, ethanol production is expected to increase three to fivefold in the future (FBP). By 2025, the Indian government has set a goal of 253 % growth, resulting in an Ethanol Rush.

### **Biomass energy: Concept of biomass energy utilization, types of biomass energy, conversion processes.**

- Biomass is biological organic matter derived from living or recently-living organisms
- Bioenergy is the energy contained (stored) in biomass
- Biomass is an extremely important energy source, available nearly everywhere



- Bioenergy is the energy retrieved from biomass sources. It is the largest used renewable energy resource in the world

## TYPES OF BIOMASS FUELS

Some of the types of biomass fuels are woody fuels, animal waste, agricultural waste, biomass crops, animal waste, dairy manure, dry animal manure, yard waste, mill residue, forest residue, landfills and alcohol fuels.

1. **Wood waste** : wood and its waste are the best source of biomass. Its combustion produces generate heat and steam which can be further used to produce electricity. All forms of wood that include wood chips, wood pallets, logs, sawdust and even tree bark can be used to produce the required energy.
2. **Animal waste** : animal waste includes animal manure, animal waste and waste from livestock. All these can be considered as raw materials for biomass. The dairy industry too contributes to a large extent.
3. **Agricultural waste** : agricultural waste is a huge source of raw material to the biomass industry. Different types of agricultural waste can be utilized to create energy. Waste can be in the form of leaves, husk, shells and animal waste. All these come under renewable energy source that are even beneficial to the environment. Another important characteristic of this agricultural waste is that it can be used for organic farming as manure and used for cooking and heating purposes.
4. **Biomass crops** : certain crops are grown so that it can be further used for biomass energy. Some of the commonly grown biomass crops are corn varieties for ethanol production and soybean oil for the production biodiesel. The heat produces can be used for generating electricity.
5. **Dairy manure** : dry manure is from livestock. The manure collected in bulk from farms are washed and dried. This is used for the

combustion in biomass plants to generate heat for the production of electricity.

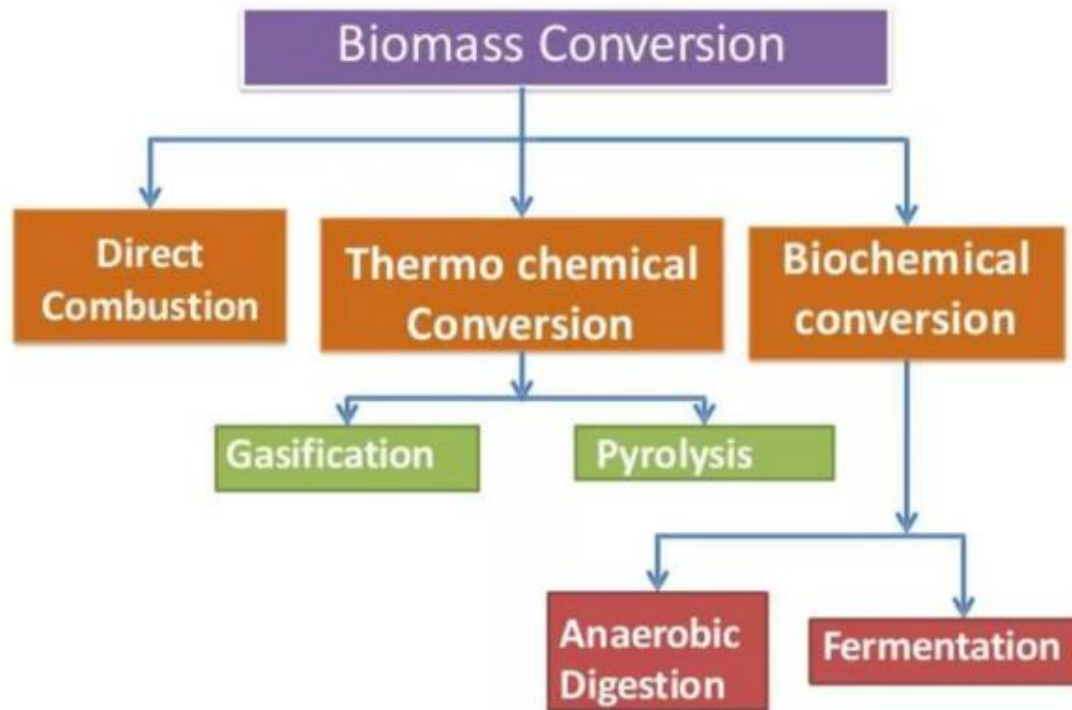
6. **Landfills** : landfills are a traditional method in which waste is buried underground so that it decomposes. So these buried wastes decompose to form biomass and produces biogas. This energy can be utilized to generate heat and energy.
7. **Mill residue** : the residue from the wood mills like wood chips and wood powder can be effectively utilized in biomass plants.
8. **Alcohol fuels** : alcohol fuel such as ethanol is produced through a process of fermentation which is widely used by automobile industry.

## METHODS OF CONVERTING BIOMASS

Biomass can be converted into the required energy form in the following ways-

1. **Combustion** : combustion is the process by which, the organic matter is burnt in the presence of oxygen which generates heat. Due to this heat water is boiled, steam is produced, turbine is turned and electricity is generated.
2. **Gasification** : gasification is a process by which organic matter is being reacted with heat to form carbon dioxide, carbon monoxide and hydrogen. Less amount of oxygen is utilized in this process. The final product is a gas named syngas (producer gas) which is also a fuel. This can also help in generating electricity and used for various other purposes.
3. **Pyrolysis** : Pyrolysis is a process by which organic matter is made into the required form at a very high temperature in the absence of oxygen. Here the organic matter does not get burnt because oxygen is not present. So it is transformed into 3 forms namely- bio oil, bio char and syngas.
4. **Bio digestion** : bio digestion is a process by which bacteria breaks down the organic matter in the absence of oxygen. This can also be termed as anaerobic respiration. The final product is biogas and combusted again to generate electricity.

5. **Fermentation** : fermentation is a process by which organic matter gets converted into alcohol with the help of yeast. The byproduct of this reaction is ethanol which is commonly used to power automobiles.



Biomass is converted to energy through various processes, including

- ❑ **Direct combustion** (burning) to produce heat
- ❑ **Thermochemical** conversion to produce solid, gaseous, and liquid fuels
- ❑ **Biological conversion** to produce liquid and gaseous fuels

#### Direct combustion

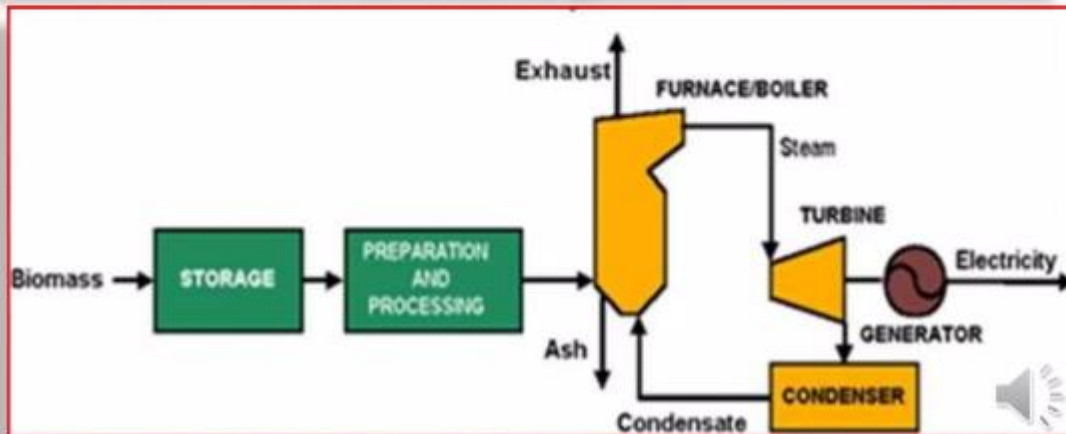
is the most common method for converting biomass to useful energy. All biomass can be burned directly for heating buildings and water, for industrial process heat, and for generating electricity in steam turbines.

#### Thermochemical conversion

of biomass includes *pyrolysis* and *gasification*. Both are thermal decomposition processes in which biomass feedstock materials are heated in closed, pressurized vessels called *gassifiers* at high temperatures. They mainly differ in the process temperatures and amount of oxygen present during the conversion process.

## Direct Combustion

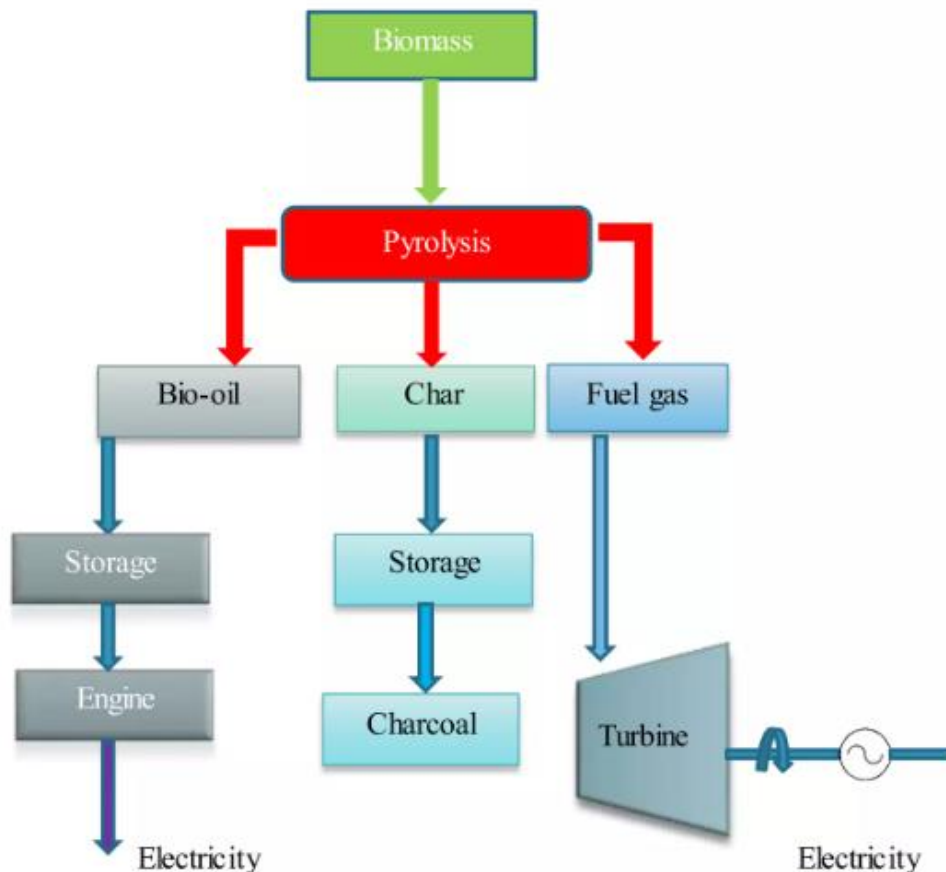
In a direct combustion system, biomass is burned in a **combustor** or **furnace** to generate hot gas, which is fed into a boiler to generate **steam**, which is expanded through a steam **turbine** or steam engine to produce **mechanical** or **electrical energy**.



- ☐ The direct combustion of biomass in **presence** of **oxygen/air** to produce heat and by products is called direct combustion.
- ☐ The complete combustion of biomass into ash is called incineration.
- ☐ This heat energy in the product gases or in the form of steam can be used for various applications like space heating or cooling, power generation process heating in industries or any other application.
- ☐ However, if biomass energy by combustion is used as co-generation with conventional fuels, the utilization of biomass energy makes it an attractive proposition.

## 1.2 Pyrolysis Process

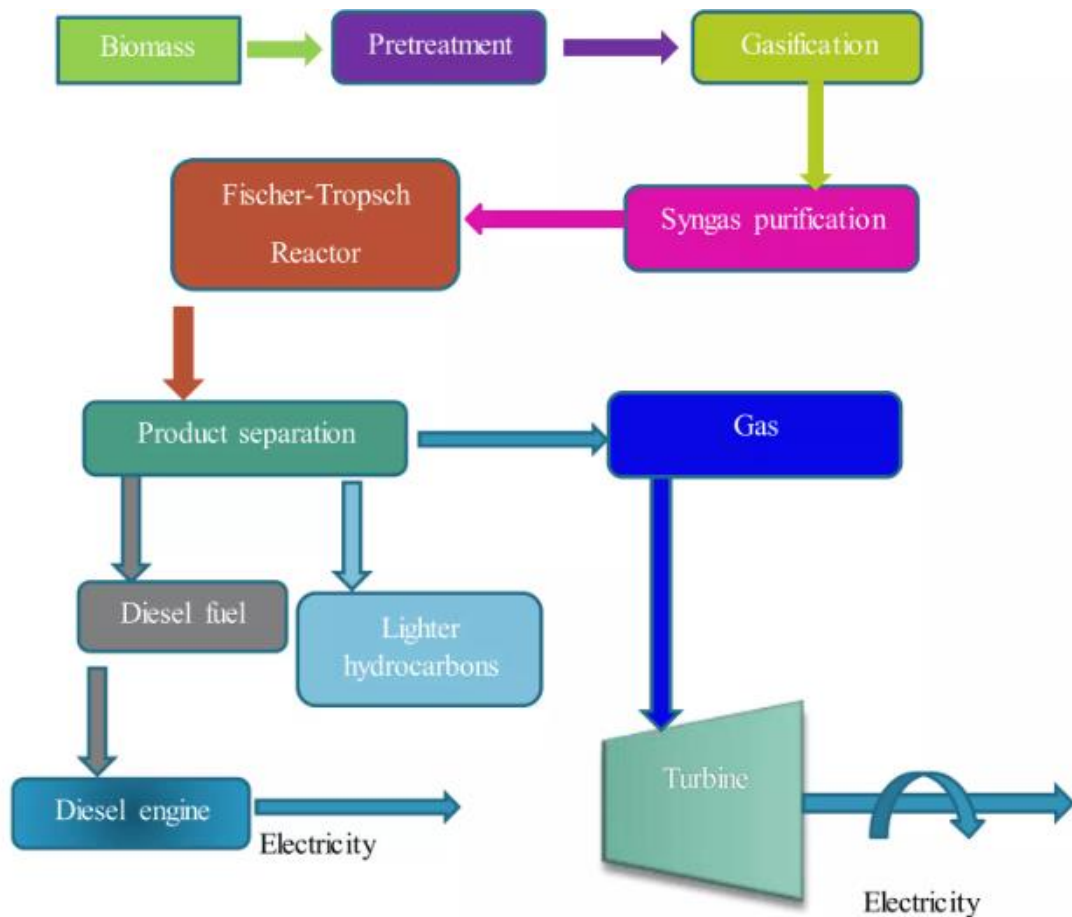
It is the process of conversion of biomass to liquid (bio-oil), solid (charcoal) and gaseous (fuel gases) products by heating in the absence of air at 500 °C. There are two types of pyrolysis: Fast pyrolysis, conventional (Carbonization) pyrolysis and slow pyrolysis. Fast pyrolysis process has high heating value and heat transfer rate and completes within seconds. Fast pyrolysis yields 60% bio-oil, 20% bio-char and 20% biogas. Conventional pyrolysis process is the process in which mostly carbon (35%) is leaved as residue. Slow pyrolysis takes more time than fast pyrolysis, it also has low temperature and heating values. Flash pyrolysis is the type of fast pyrolysis, in which 80% bio-oil is obtained at keeping temperature low. If flash pyrolysis is used for converting biomass to bio-crude, it has up to 80% efficiency [15][16].





### 1.3 Gasification process

In biomass gasification, charcoal, wood chips, energy crops, forestry residues, agricultural waste and other wastes are transformed into flammable gases at high temperature (800-1000°C). In this process fuel (biomass) reacts with a gasifying medium such as oxygen enriched air, pure oxygen, steam or a combination of both. The product gas composition and energy content depends upon the gasifying media's nature and amount of it. Low calorific Value (CV) gas obtained by gasification about 4-6 MJ/N m<sup>3</sup>. The product gas can be used as a feedstock (syngas) in the production of chemicals like methanol. One promising concept is the biomass integrated gasification/ combined cycle (BIG/CC), in which gas turbines convert the gaseous fuel to electricity with a high overall conversion efficiency. The integration of gasification and combustion/ heat recovery ensures 40-50% conversion efficiency for a 30-60 MW. The syngas can be converted into hydrogen gas, and it may have a future as fuel for transportation [15].

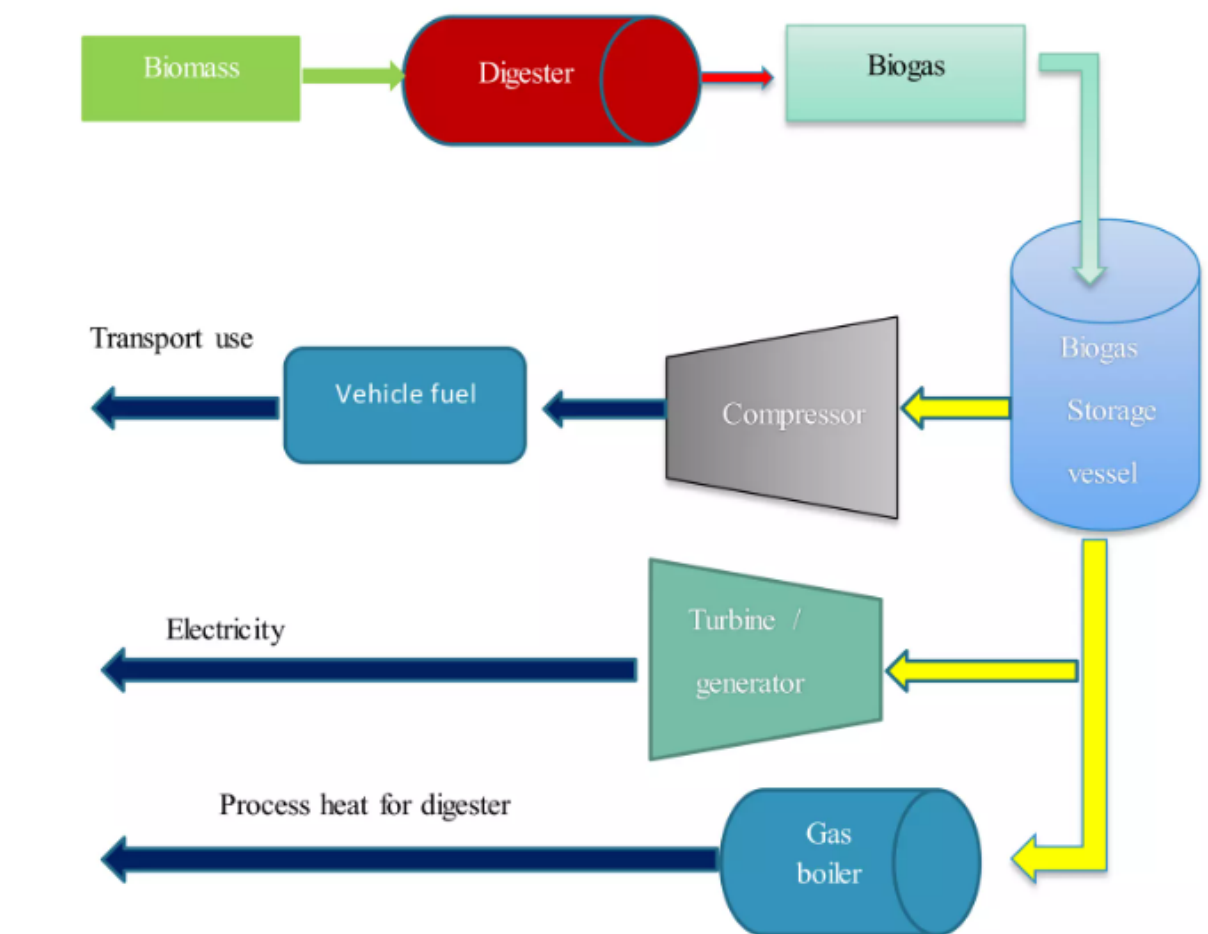


## 2. Bio-Chemical conversion

Biochemical conversion makes use of the enzymes of bacteria and other living organisms to break down biomass and convert it into fuels. This conversion process includes anaerobic digestion and fermentation.

### 2.1. Anaerobic digestion process

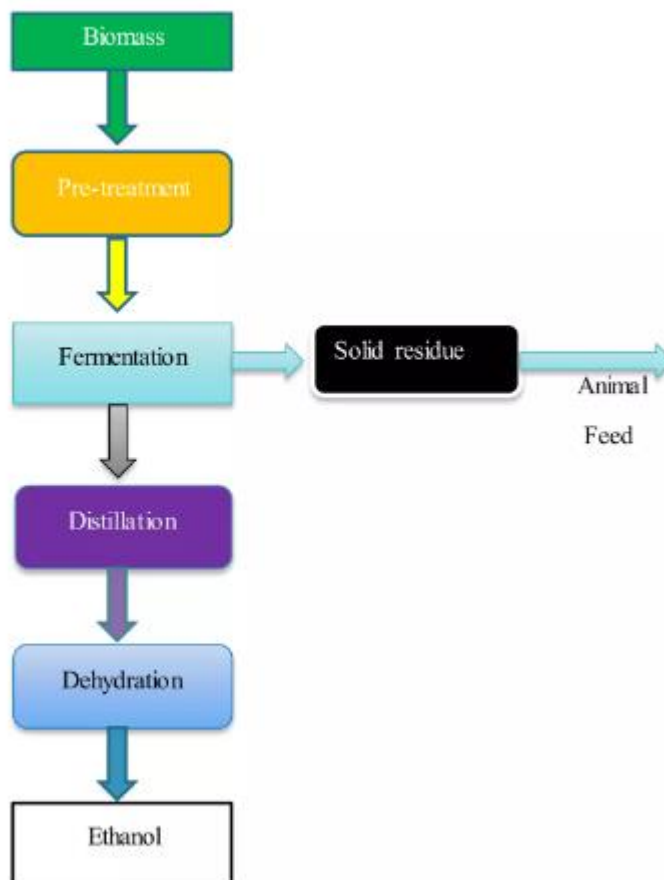
This is a process in which organic material directly converted to a gas which is termed as biogas. It is mixture of methane, carbon dioxide and other gases like hydrogen sulphide in small quantities. Biomass is converted in anaerobic environment by bacteria, which produces a gas having an energy of 20-40% of lower heating value of the feedstock. This process is suitable for organic wastes having high moisture about 80-90%. This biogas can be directly used in spark ignition gas engines and gas turbines and can be upgraded to higher quality natural gas by removing carbon dioxide. The overall conversion efficiency of this process is 21%. Waste heat from engines and turbines can be recovered by using combined heat and power system [17].





## 2.2. Fermentation process

Fermentation is an anaerobic process that breaks down the glucose within organic materials. It is a series of chemical reactions that convert sugars to ethanol. The basic fermentation process involves the conversion of a plant's glucose (or carbohydrate) into an alcohol or acid. Yeast or bacteria are added to the biomass material, which feed on the sugars to produce ethanol and carbon dioxide. The ethanol is distilled and dehydrated to obtain a higher concentration of alcohol to achieve the required purity for the use as automotive fuel. The solid residue from the fermentation process can be used as cattle-feed and in the case of sugar cane; the bagasse can be used as a fuel for boilers or for subsequent gasification [18].



## ADVANTAGES OF USING BIOMASS

- **Renewable source** : biomass is considered as a renewable source of energy as compared to the other forms of energy. This is mainly because of the raw materials that is used as which is available throughout. Its procurement and regrowth is easy. [Also read about [renewable vs. non-renewable energy](#)]
- **Cheaper** : the production of biomass energy is comparatively cheaper as compared to fossil fuel. The raw materials are cheaply

available. So the low cost in the generation of electricity reduces the bills of the common man. This makes biomass energy more attractive.

- **Variety products** : biomass energy is versatile as it creates a lot of products. Biomass can be converted into various forms in the presence and absence of oxygen. Some of the byproducts are ethanol, biogas, syngas, bio-oil and bio-char.
- **Clean gas** : biomass energy is a cleaner gas as compared to other forms of energy. Green house gases are not emitted during combustion of organic matter. Minimal [pollution](#) is outcome. Lesser amount of carbon is emitted during the process which is absorbed by plants for their survival and life cycle.

## DISADVANTAGES OF USING BIOMASS

- Constant and continues supply of biomass is required for the generation of biomass energy.
- Compared to the input of raw materials, the outcome is comparatively less.
- All the raw materials used are waste products that can cause pollution and foul smell.
- The storage of biogas and its transportation is difficult due to less advanced technology.
- Large space is required for the building of plants.
- Huge investment is required for biomass plants.
- More and more biomass crops are grown which in turn reduces the fertility of the soil.